

Electric Logistics Vehicle Route Planning with Time Windows Based on Improved Genetic Algorithm

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Abstract

The widespread application of new energy electric vehicles in the logistics field has promoted the research and development of electric vehicle path issues (EVRP). In view of the Electric Vehicle Routing Problem with Time Windows (EVRPTW), this paper establishes an optimization model with the goal of minimizing logistics and distribution costs, and comprehensively considers various factors such as transportation cost, time window penalty cost, and charging cost. By incorporating Tabu Search Hill Climbing into the Genetic Algorithm (GA) for enhanced performance, the global search ability and local optimization effect of the algorithm are significantly improved. The test of the example data set shows that the optimization effect of the improved algorithm has been improved by 8.75%, which provides a practical solution for logistics and distribution path planning.

CCS Concepts

• **Computing methodologies** → Artificial intelligence; Control methods; Motion path planning.

Keywords

EVRPTW, Distribution costs, Tabu Search Hill Climbing, GA

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1 INTRODUCTION

In recent years, with the intensification of the energy crisis and environmental problems, new energy electric vehicles (EVs) are gradually replacing traditional fuel vehicles and becoming a new trend for logistics companies [1]. This makes the problem of electric vehicle routing a key challenge for traditional logistics companies in the case of limited charging stations (CSs).

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Vehicle Routing Problem (VRP) was first proposed by Dantzig and Ramser in 1959 [2]. After years of development, many definitions of different problem variants have been derived in practical optimization problems [3], among which Electric Vehicle Routing Problem with Time Windows (EVRPTW) stands out as a significant variant. EVRPTW is based on VRP, combines the charging and range limits of EVs, and takes into account the constraints of customer service time windows. It increases the complexity and reality of the problem and is considered an NP-hard problem, which is difficult to solve efficiently and accurately through precise algorithms [4], while heuristic algorithms are considered to be an effective method to solve such NP-hard problems. Therefore, this paper uses Genetic Algorithm (GA), one of the modern heuristic algorithms, to solve EVRPTW in logistics transportation. However, traditional GA has poor quality in the initial population and is prone to getting stuck in local optimal solutions when the solution space is complex. To address this, this paper proposes an improved GA to optimize the EVRPTW.

2 THE MATHEMATICAL MODEL OF THE EVRPTW

This paper takes the minimization of total distribution cost as the objective function and comprehensively considers fixed costs, transportation costs, time window penalty costs, charging costs and opportunity costs. Establish multiple constraints, including: service time window constraints, maximum mileage constraints, volume constraints, etc.

2.1 Basic Assumption

To simplify the optimization model for urban logistics transportation, the following assumptions need to be met:

(1) Distribution model assumption: This model consists of a single distribution center including CSs and multiple distribution points. Distribution centers and customer locations are known.

(2) Distribution point assumption: The demand for goods, service time window and geographical location of each distribution point are all known, and the demand at the distribution point cannot be divided. The demand for goods is less than the approved load capacity and volume of the vehicle. EVs can serve multiple distribution points within the specified service time window, but each distribution point can only be served once.

(3) Line assumption: All EVs depart from the distribution center in a fully-charged state to complete the corresponding distribution point tasks and return without proceeding to the next task.

Table 1: Model parameters.

Parameter	Definition	Parameter	Definition
$V = \{0\} \cup N$	Collection of all nodes	d_{ij}	Distance from node i to j
0	Distribution center or CS	r_1	Range efficiency
N	Delivery point collection	r_2	Charging rate
K	Vehicle collection	x_{ij}^k	Binary variable, whether vehicle k is from node i to j
F	Fixed cost	f_i	The volume of goods at distribution point i
M	Cruising range	w_i	The weight of the goods at distribution point i
Q	Payload volume	$[E_i, L_i]$	The window of service that distribution point i wishes to arrive
W	Payload capacity	$[e_i, l_i]$	The maximum service time window acceptable to distribution point i
c_1	Transportation cost per unit distance	t_i^k	The time when vehicle k arrives at node i
c_2	Unit electricity price	α	Early arrival penalty cost
c_3	Opportunity cost per unit time when charging	β	Late arrival penalty costs

(4) Energy assumption: The power consumption and charging time of EVs are linear and proportional to the driving mileage, regardless of loading and unloading time.

(5) Cost Assumptions: Charging costs, opportunity costs, penalty costs, and fixed costs per vehicle are all known. The penalty cost for arriving early in the service time window is higher than the penalty cost for arriving late.

2.2 Parameter Description

The relevant symbol explanations of the objective function and distribution points are shown in Table 1.

2.3 Objective function

(1) Fixed cost: Mainly include labor costs, EVs maintenance costs, depreciation costs, management costs, driver wages, etc. These costs are mainly related to the number of EVs used. As shown in formula (1):

$$C_1 = F \sum_{j \in N} \sum_{k \in K} x_{0j}^k \quad (1)$$

(2) Transportation cost: Mainly includes the expenses incurred by EVs during driving, as shown in equation (2):

$$C_2 = c_1 \sum_{i,j \in V} \sum_{k \in K} d_{ij} x_{ij}^k \quad (2)$$

(3) Penalty cost: Mainly based on time window constraints. When EVs arrive within the $[E_i, L_i]$ standard service time window interval, no penalty fee will be incurred; when EVs arrive within the $[e_i, E_i]$ time interval, a smaller penalty will be incurred Cost; when arriving within the $[L_i, l_i]$ time interval, a large penalty fee R will be generated; if it arrives at other times, a huge penalty fee R will be generated, as shown in equation (3):

$$C_3 = \sum_{i \in V} \sum_{k \in K} \begin{cases} R, & t_i^k < e_i \text{ or } t_i^k > l_i \\ \alpha \cdot \max(0, E_i - t_i^k), & t_i^k \in [e_i, E_i] \\ \beta \cdot \max(0, t_i^k - L_i), & t_i^k \in [L_i, l_i] \end{cases} \quad (3)$$

(4) Charging cost: Mainly related to the electricity consumed by EVs, as shown in equation (4):

$$C_4 = c_2 \sum_{i,j \in V} \sum_{k \in K} x_{ij}^k d_{ij} / r_1 \quad (4)$$

(5) Opportunity cost: Mainly related to the charging time of EVs, as shown in equation (5):

$$C_5 = c_3 \sum_{i,j \in V} \sum_{k \in K} x_{ij}^k d_{ij} / (r_1 \cdot r_2) \quad (5)$$

In summary, the total delivery cost is expressed as shown in Equation (6):

$$C = \sum_{i=1}^5 C_i \quad (6)$$

2.4 Constraints

In VRPTW, the role of constraints is to ensure that the path plan achieves the optimization goal and improves the solution efficiency while meeting the actual operating constraints. The formula is as follows:

$$\sum_{i,j \in V} \sum_{k \in K} x_{ij}^k d_{ij} \leq M, \forall i \neq j \quad (7)$$

$$\sum_{i,j \in V} \sum_{k \in K} x_{ij}^k f_i \leq Q, \forall i \neq j \quad (8)$$

$$\sum_{i,j \in V} \sum_{k \in K} x_{ij}^k w_i \leq W, \forall i \neq j \quad (9)$$

$$\sum_{i \in V} \sum_{j \in N} \sum_{k \in K} x_{ij}^k = \sum_{i \in V} \sum_{j \in N} \sum_{k \in K} x_{ji}^k = 1, \forall i \neq j \quad (10)$$

$$\sum_{j \in N} \sum_{k \in K} x_{0j}^k = \sum_{j \in N} \sum_{k \in K} x_{j0}^k \leq 1, \forall i \neq j \quad (11)$$

$$\sum_{i \in V} \sum_{j \in N} x_{ij}^k \leq |K|, \forall i \neq j \quad (12)$$

$$e_i \leq t_i^k \leq l_i \quad (13)$$

(7) represents the maximum mileage constraint of EV; (8) represents the maximum volume constraint of EV; (9) represents the

maximum load constraint of EV; (10) represents that each distribution point can only be served by one EV once; (11) Represents the path constraint of EV, each vehicle can only start from the distribution center, and finally returns to the distribution center; (12) represents the constraint on the number of distribution EVs; (13) represents the service time window constraint.

3 IMPROVED GENETIC ALGORITHM

3.1 Chromosome encoding and decoding

The chromosome structure is encoded using natural numbers, where 0 represents the distribution center. K EVs start from the distribution center, deliver to distribution points numbered 1, 2, ..., n , and then return to the distribution center. The chromosome length for a distribution route is $n+k+1$. For example, a chromosome {5,1,3,2,4} represents a feasible solution with a total of 5 distribution points, where each number corresponds to a distribution point.

When the chromosome is decoded, EV arrangements are carried out according to constraints such as payload capacity. If the total demand at distribution points 5, 1, and 3 exceeds the EV's payload capacity, but the sum of the demands at distribution points 5 and 1 is within the EV's capacity, the first distribution route is $0 \rightarrow 5 \rightarrow 1 \rightarrow 0$, where 0 represents the distribution center. Similarly, if the sum of the demands at distribution points 3, 2, and 4 is within the EV's capacity, the second distribution route is $0 \rightarrow 3 \rightarrow 2 \rightarrow 4 \rightarrow 0$. Thus, the complete distribution scheme is {0,5,1,0,3,2,4,0}. This means that the first EV starts from the distribution center, serves distribution points 5 and 1, and then returns to the distribution center. The second EV starts from the distribution center, serves distribution points 3, 2, and 4, and then returns to the distribution center.

3.2 Population Initialization

During population initialization, Collection and Fisher-Yates algorithm are used to generate all individuals to ensure the diversity and uniform distribution of the population and avoid the repeatability and deviation of the initial solution.

3.3 Determine the fitness function

The fitness function evaluates the quality of an individual in an optimization problem. It reflects its performance in the problem by assigning a fitness value to each solution, and determines the probability of the individual being passed on to the next generation. This paper uses the reciprocal of the objective function as the individual fitness value. The higher the fitness value, the better the individual, and the greater the probability of inheriting it to the next generation. As shown in equation (14).

$$fitness(i) = \frac{1}{C} \quad (14)$$

3.4 Genetic operator design

The Elite Retention Strategy is introduced before the selection operation to ensure that individuals with high performance directly enter the offspring to prevent them from being lost during the evolution process.

(1) The selection operator employs Roulette Wheel Selection, where the probability of selecting an individual is proportional to its fitness value.

(2) The crossover operator uses Partially Matched Crossover (PMX) to ensure that genes in the chromosome appear only once and will not be repeated.

(3) Mutation operators use Swap Mutation, randomly selecting two different positions on the chromosome for gene exchange.

3.5 Tabu Search Hill Climbing

Tabu Search Hill Climbing is an optimization approach that combines the Tabu Search algorithm with the Hill Climbing method. Its core lies in enhancing the classical Hill Climbing algorithm by maintaining a tabu list that records visited solutions or their characteristics, thus avoiding redundant searches and getting trapped in local optima. Following the mutation operation in GA, Tabu Search Hill Climbing is introduced to perform local optimization on the mutated individuals through neighborhood search, further improving their fitness. It also utilizes the tabu list to restrict repetitive paths, aiding in escaping local optima and expanding the search scope. This method balances local optimization and global search, not only refining the quality of mutated individuals but also preventing the search from being confined to local optima, thereby significantly enhancing the overall performance and search efficiency of the population.

4 EXAMPLE ANALYSIS AND DISCUSSION

This experiment employs Python programming, running on a laptop equipped with an Intel (R) Core (TM) i5-7300HQ CPU at 2.50GHz and 16.0GB of memory. The dataset for the case study originates from the field survey conducted by Song Jianghao [5]. It consists of 33 nodes, where node 0 represents the distribution center, and nodes 1 to 32 represent delivery points. The node distribution is shown in Figure 1, and the relevant parameters of the EVs at the distribution center are listed in Table 2.

4.1 Experimental Conclusion

In this experiment, it is assumed that EVs depart from the distribution center for delivery at 1:30 PM. The fixed cost of the EVs is set at 200 CNY/day, the charging cost is 0.2 CNY/min, and the opportunity cost is 0.1 CNY/min. The charging rate at the charging station (CS) is 40 kW. The penalty cost for early arrival is 2 CNY/min, while the penalty cost for late arrival is 5 CNY/min.

The population size, elite retention probability, crossover probability, mutation probability, and number of iterations for the GA are set to 100, 0.1, 0.8, 0.1, and 500, respectively. For the Tabu Search Hill Climbing algorithm, both the tabu list length and the number of iterations are set to 10.

The improved method proposed in this paper was compared with the optimization strategy developed by Song Jianghao [5], and the results showed that the number of vehicles used was the same at 13. However, the objective function values were 2943.59 CNY and 3225.96 CNY, respectively, reflecting a total distribution cost reduction of 8.75%. These results demonstrate that the improved GA proposed in this paper achieves better optimization performance for the objective function. The comparison of algorithm iteration curves is shown in Figure 2.

The 13 vehicle delivery routes corresponding to the optimal objective function value obtained by the improved GA proposed in

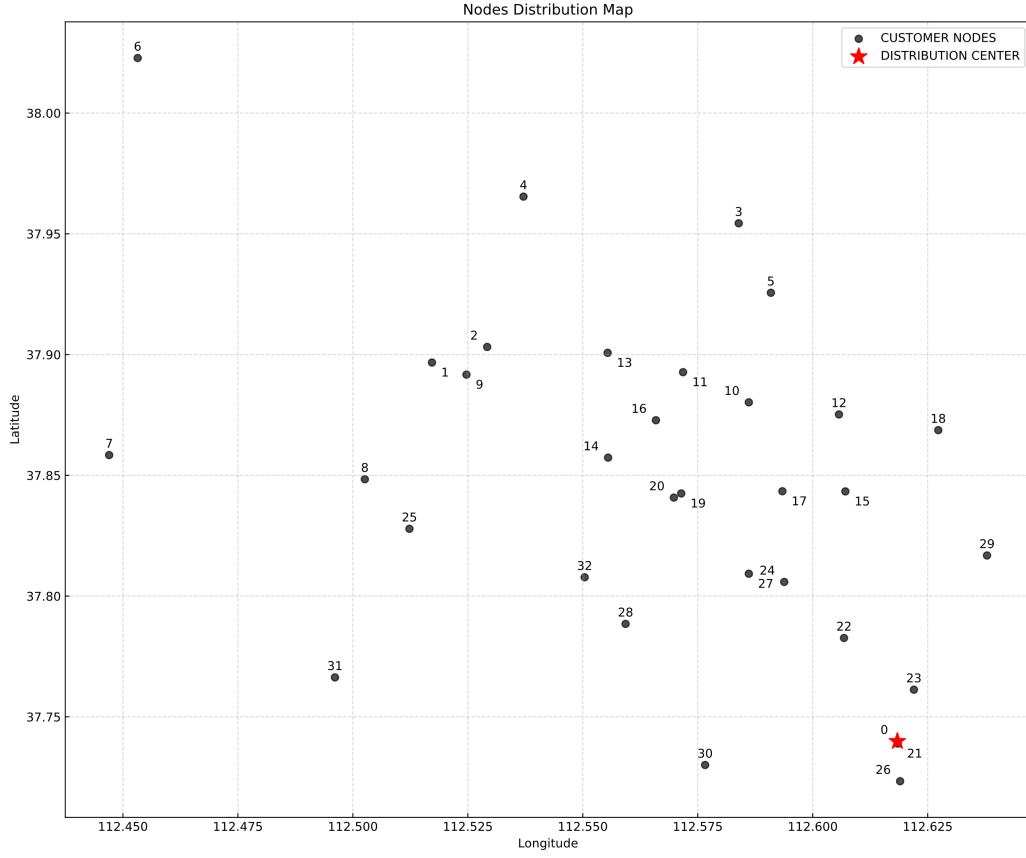


Figure 1: Node distribution map.

Table 2: EVs parameters.

Payload capacity	Payload volume	Cruising range	Range efficiency	Transportation cost	Number of vehicles
2t	12.763842 m ³	200km	1km/Kwh	0.5CNY/km	35

this paper are shown in Table 3. Each delivery route is served by a separate vehicle (see Figure 3).

Each delivery route is served by a separate vehicle (see Table 3). And the total cargo volume of the delivery points on each route does not exceed the relevant constraints of the vehicles.

5 CONCLUSIONS AND OUTLOOK

This paper addresses the optimization of distribution paths for electric logistics vehicles, based on the requirements of the distribution center. It constructs an optimization model with total distribution cost as the objective, employs an improved genetic algorithm to solve the model, and verifies the effectiveness of the results through comparative analysis. The findings provide a practical basis for optimizing logistics distribution routes.

However, this study has several limitations. It focuses on a single-objective optimization problem, while real-life scenarios often require multi-objective optimization to balance various factors.

Table 3: Distribution Route.

Vehicle 1	0→9→1→2→0
Vehicle 2	0→18→12→27→0
Vehicle 3	0→25→24→0
Vehicle 4	0→32→6→28→0
Vehicle 5	0→19→20→0
Vehicle 6	0→22→29→0
Vehicle 7	0→26→21→0
Vehicle 8	0→17→15→0
Vehicle 9	0→14→16→23→0
Vehicle 10	0→30→31→0
Vehicle 11	0→11→13→5→0
Vehicle 12	0→7→8→0
Vehicle 13	0→10→3→4→0

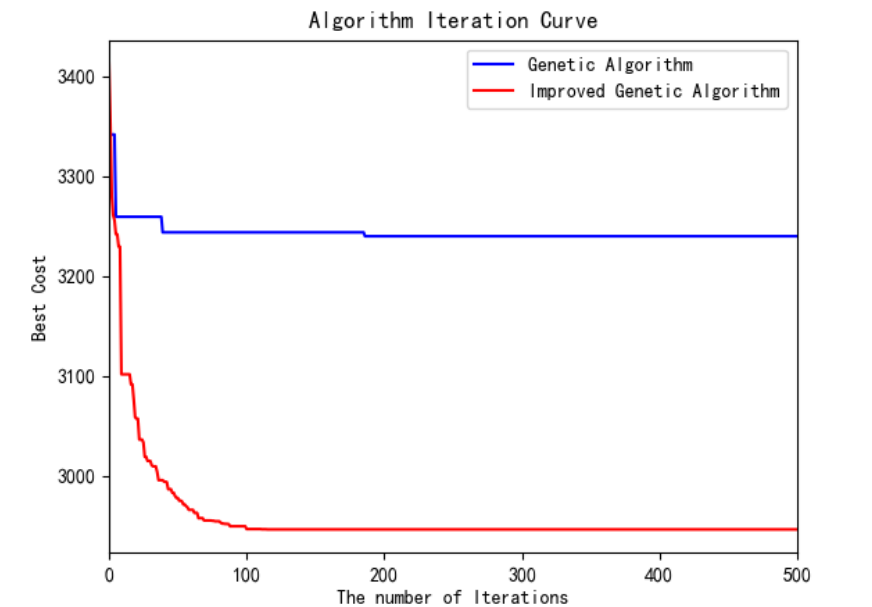


Figure 2: Algorithm Iteration Curve.

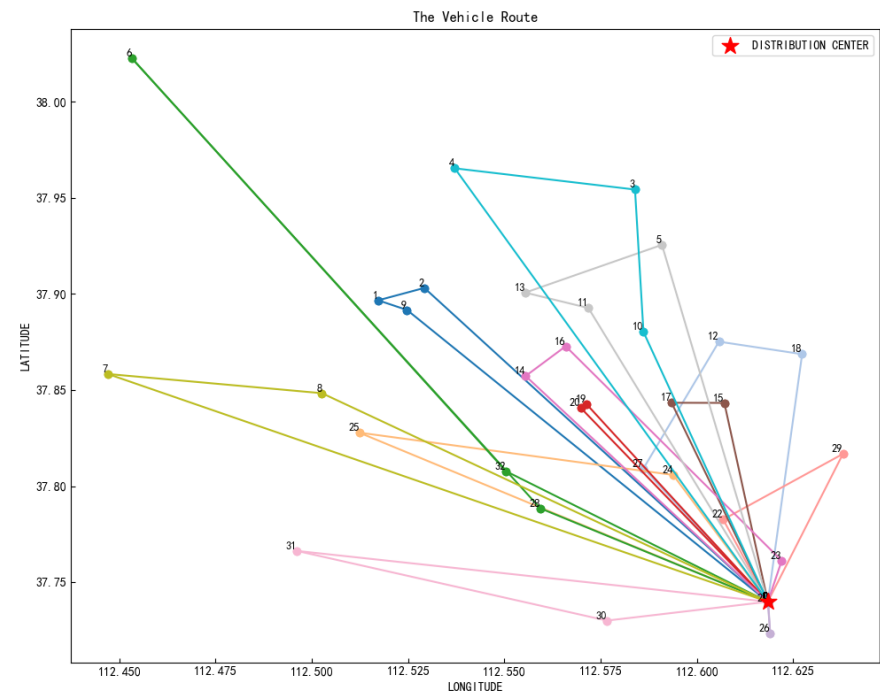


Figure 3: The Vehicle Route.

Additionally, all vehicles in this study can only be charged at the distribution center, without considering en-route charging. Future research could incorporate multiple charging stations and account for loading and unloading times to further refine logistics vehicle route planning.

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